

Wave 5.2B Offset And Harmonic Guided Campaign Results Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report closes out the completed `wave 5.2B` offset and harmonic guided campaign on `polished_dataset`.

Campaign evidence

- Campaign: `wave52b_offset_harmonic_guided_campaign_2026_07_01`
- Dataset: `polished_dataset`
- Dataset schema: `polished_point_v1`
- Campaign output:
`output/training_campaigns/2026-07-01-18-59-04_wave52b_offset_harmonic_guided_campaign_2026_07_01/`
- Completed runs: `12`
- Failed runs: `0`
- Generated at: `2026-07-02T02:27:12`

The campaign tested four lightweight profiles across `global`, `Fw`, and `Bw` surfaces:

- `pointwise_control`
- `offset_head`
- `offset_centered_shape`
- `offset_centered_shape_harmonic`

This is a normal scalar training closeout. It does not run or replace the official `TE Curve Verification Pipeline`, and it does not make a curve-first promotion decision.

Scalar Winner

The scalar campaign winner is

Field	Value
Run name	<code>te_wave52b_offset_harmonic_guided_offset_centered_shape_harmonic_fw</code>
Run instance	<code>2026-07-02-01-24-47__te_wave52b_offset_harmonic_guided_offset_centered_shape_harmonic_fw</code>
Model family	<code>wave52b_offset_harmonic_guided_offset_centered_shape_harmonic_fw</code>
Surface	<code>Fw</code>
Test MAE [deg]	<code>0.001391538535244763</code>
Test RMSE [deg]	<code>0.0017712278058752418</code>
Val MAE [deg]	<code>0.001809177570976317</code>
Parameters	<code>22593</code>
Checkpoint	<code>output\training_runs\wave52b_offset_harmonic_guided_offset_centered_shape_harmonic_fw\2026-07-02-01-24-47__te_wave52b_offset_harmonic_guided_offset_centered_shape_harmonic_fw\checkpoints\wave52b_offset_harmonic_guided-epoch=116-val_mae=0.00180918.ckpt</code>

The winner is the forward-surface harmonic profile, not a global model.

Surface Winners

Surface	Best Profile	Best Run	Test MAE [deg]	Test RMSE [deg]	Val MAE [deg]
global	offset_centered_shape_harmonic	te_wave52b_offset_harmonic_guided_offset_centered_shape_harmonic_global	0.002215	0.002799	0.001886
Fw	offset_centered_shape_harmonic	te_wave52b_offset_harmonic_guided_offset_centered_shape_harmonic_fw	0.001392	0.001771	0.001809
Bw	offset_centered_shape_harmonic	te_wave52b_offset_harmonic_guided_offset_centered_shape_harmonic_bw	0.001677	0.002151	0.002320

The same profile wins all three surfaces. That is the main technical result of the campaign: the explicit combination of offset, centered-shape, and harmonic guidance is consistently better than the control profiles inside this lightweight family.

Full Scalar Leaderboard

Rank	Run	Surface	Test MAE [deg]	Test RMSE [deg]	Val MAE [deg]
1	offset_centered_shape_harmonic_fw	Fw	0.001392	0.001771	0.001809
2	offset_centered_shape_harmonic_bw	Bw	0.001677	0.002151	0.002320
3	offset_centered_shape_fw	Fw	0.001931	0.002445	0.002258
4	offset_head_fw	Fw	0.001948	0.002454	0.002256
5	pointwise_control_bw	Bw	0.001979	0.002587	0.002591
6	offset_head_bw	Bw	0.002008	0.002632	0.002597
7	offset_centered_shape_bw	Bw	0.002012	0.002626	0.002604
8	pointwise_control_fw	Fw	0.002054	0.002564	0.002344
9	offset_centered_shape_harmonic_global	global	0.002215	0.002799	0.001886
10	pointwise_control_global	global	0.002461	0.003142	0.002210
11	offset_head_global	global	0.002483	0.003166	0.002249
12	offset_centered_shape_global	global	0.002540	0.003229	0.002271

Ablation Interpretation

The internal ablation is clear

- offset_centered_shape_harmonic is the strongest profile on every surface.
- The harmonic branch provides the largest forward gain, reducing forward test MAE from the pointwise control value of 0.002054 to 0.001392 .
- The backward gain is also material, reducing backward test MAE from 0.001979 to 0.001677 .
- The global surface improves relative to all other Wave 5.2B global profiles, but it remains weaker than the best polished early-wave global models. This supports keeping harmonic structure in the candidate ingredient list for later wave 5.2C or wave 6 integration. It does not prove that this lightweight model family should replace the stronger sequence backbones.

Comparison Against Current Polished Leaders

The current polished early-wave scalar leaders remain stronger:

Surface	Polished Leader	Leader MAE	W5.2B Best	W5.2B MAE
global	te_periodic_lstm_sequence_global	0.001187	offset_centered_shape_harmonic_global	0.002215
Fw	te_periodic_gru_sequence_fw	0.001101	offset_centered_shape_harmonic_fw	0.001392
Bw	te_periodic_gru_sequence_bw	0.001084	offset_centered_shape_harmonic_bw	0.001677

Therefore this campaign does not change the scalar program winner, which remains `te_periodic_gru_sequence_bw`.

Registry And Status Decision

The run artifacts and family registries are accepted as completed training evidence for `Wave 5.2B`.

Closeout decisions

- Accept the `12` completed runs and their campaign leaderboard.
- Record `offset_centered_shape_harmonic` as the strongest Wave 5.2B profile.
- Do not promote Wave 5.2B over the polished early-wave scalar leaders.
- Do not alter official `TE Curve Verification Pipeline` accepted leaders in this closeout.
- Keep the externally running `polished_dataset_full_wave_retraining_2026_06_22` campaign isolated until its final artifacts are synchronized on its own workstation.

Next Steps

Recommended next actions

1. Keep Wave 5.2B as evidence that harmonic guidance should be retained as an integration ingredient.
2. Treat an official `TE Curve Verification Pipeline` refresh for Wave 5.2B as a separate optional operator-approved step.
3. Wait for the externally running full-wave polished retraining campaign before drawing final clean-branch conclusions.
4. Use Wave 5.2B results to inform `Wave 5.2C` dirty-to-clean or transfer supervision only after curve-first evidence is available or explicitly deferred.